

Tissue perfusion pressure: A novel hemodynamic measure to assess risk of acute kidney injury after cardiac surgery



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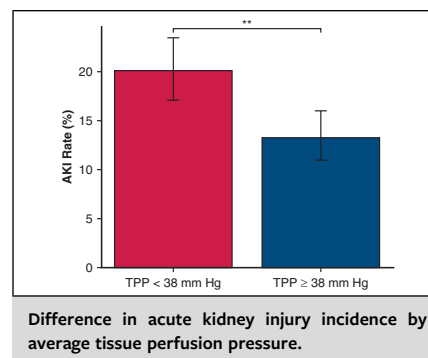
ABSTRACT

Background: Optimal hemodynamic targets for preventing acute kidney injury (AKI) have remained elusive. We hypothesized that lower tissue perfusion pressure (TPP), a novel perfusion index representing the difference between mean arterial pressure and the critical closing pressure, is predictive of AKI after cardiac surgery.

Methods: Individual patient TPP waveforms were constructed from continuous hemodynamic data in 1224 patients after cardiac surgery. The relationship of TPP and AKI was determined and stratified by vasoactive inotrope score. Logistic regression was performed to identify the optimal TPP threshold for predicting AKI. Unsupervised machine learning was used to explore different hemodynamic phenotypes and their association with AKI.

Results: AKI occurred in 17.6% of patients and was associated with higher rates of mortality (15.8% vs 2.0%; $P < .001$) and major morbidity (45.1% vs 14.2%; $P < .001$) and significantly lower average TPP (37.6 mm Hg [33.7–41.0 mm Hg] vs 39.0 mm Hg [34.6–42.6 mm Hg], $P < .001$). A threshold TPP < 38 mm Hg effectively stratified patients by AKI risk (odds ratio, 1.75; 95% CI, 1.30–2.35). For patients requiring vasoactive medications, average TPP < 38 mm Hg indicated higher risk of AKI independent of average mean arterial pressure (adjusted odds ratio, 1.69; 95% CI, 1.17–2.45). K-means clustering identified a high-risk phenotype with lower average TPP (35.6 mm Hg [30.9–39.8 mm Hg] vs 40.4 mm Hg [35.9–44.5 mm Hg]; $P < .001$), higher average vasoactive inotrope score (2.8 [0.3–6.8] vs 0.5 [0.0–2.5], $P < .001$), and greater incidence of AKI (29.8% vs 10.1%; $P < .001$).

Conclusions: Lower TPP is associated with greater risk of AKI after cardiac surgery. TPP could serve as an adjunct to traditional hemodynamic measures to guide hemodynamic management, especially in patients with higher vasopressor requirements. (J Thorac Cardiovasc Surg 2026;171:455–62)



CENTRAL MESSAGE

Lower tissue perfusion pressure in the early postoperative period after cardiac surgery is associated with an elevated risk of acute kidney injury.

PERSPECTIVE

TPP is a novel hemodynamic measure that can be estimated from routinely collected continuous hemodynamic data. TPP < 38 mm Hg was associated with a higher risk of acute kidney injury after cardiac surgery. TPP could serve as an additional hemodynamic measure to assess renal perfusion, particularly in patients on vasopressors.

See Commentary on page 463.

Acute kidney injury (AKI) is a frequent, morbid, and costly complication of cardiac surgery, occurring in approximately 20% of cases and contributing to both short- and long-term

mortality.^{1–3} To date, no pharmacologic therapy has been shown to reduce or prevent AKI after cardiac surgery.^{4,5} Rather, given that renal hypoperfusion is a major driver of

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Abbreviations and Acronyms

AKI	= acute kidney injury
CVP	= central venous pressure
DPP	= diastolic perfusion pressure
EMR	= electronic medical record
KDIGO	= Kidney Disease: Improving Global Outcomes
MAP	= mean arterial pressure
MPP	= mean perfusion pressure
P _{crit}	= critical closing pressure
STS	= Society of Thoracic Surgeons
TPP	= tissue perfusion pressure
VIS	= vasoactive inotrope score



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postoperative AKI, optimizing hemodynamics perioperatively remains a central strategy for mitigating renal injury.^{6,7}

Several hemodynamic parameters, including mean arterial pressure (MAP), mean perfusion pressure (MPP), and diastolic perfusion pressure (DPP), are commonly used as surrogate indices of renal perfusion to guide perioperative hemodynamic management. However, their clinical application remains inconsistent, with no consensus on optimal target thresholds for mitigating AKI.⁸⁻¹⁰ Randomized trials have yielded conflicting evidence regarding whether higher MAP targets confer renal protection after cardiac surgery.¹¹⁻¹³ This uncertainty has prompted interest in alternative, data-driven perfusion indices.

Tissue perfusion pressure (TPP), defined as the difference between MAP and the critical closing pressure (P_{crit}; that is, the MAP below which blood flow ceases), may provide a more direct and individualized estimate of tissue perfusion.¹⁴⁻¹⁶ Recent work suggests that TPP can be estimated continuously from routinely collected electronic medical record (EMR) data and is associated with mortality after cardiac surgery.¹⁷ However, its utility as a measure of renal perfusion remains uncharacterized.

In this retrospective analysis, we computed postoperative TPP in a large cohort of patients who underwent cardiac surgery and investigated its association with AKI. Then, a target TPP value that best stratifies patients by their risk of subsequent AKI was identified. Lastly, the influence of vasoactive medications on TPP and AKI was evaluated, and distinct hemodynamic phenotypes were identified with unsupervised machine learning.

METHODS**Study Cohort**

Adults aged 18 years or older who underwent cardiac surgery with cardiopulmonary bypass at Baylor St Luke's Medical Center from January 1, 2021, to December 31, 2022, were identified in the Baylor College of Medicine institutional Society of Thoracic Surgeons (STS) Database. Patients who required dialysis preoperatively, had chronic kidney disease (baseline creatinine level ≥ 4.0 mg/dL), lacked baseline creatinine values, had received an organ transplant, or required mechanical circulatory support were excluded. This study was approved by the Baylor College of Medicine Institutional Review Board with waiver of informed consent (H-44702; March 23, 2021).

Calculation of Perfusion Indices

Time-series hemodynamic data including MAP, systolic and diastolic blood pressure, and central venous pressure (CVP) obtained over the first 24 hours after surgery were resampled into 5-minute increments by using means (Appendix E1). The MPP was calculated as the difference between MAP and CVP; the DPP was defined as the difference between diastolic blood pressure and CVP. The vasoactive inotrope score (VIS) was used to quantify total dose of vasopressors and inotropes in the 24 hours after surgery (Table E1).

The postoperative TPP was estimated according to the protocol described by Chandrasekhar and colleagues.¹⁷ Briefly, TPP was calculated iteratively over 5-minute intervals as the MAP minus the P_{crit}, which was computed by extrapolating the y-intercept of the linear regression of cardiac output against MAP; cardiac output was approximated from the product of pulse pressure and heart rate (Figure E1). To ensure the robustness of the estimated P_{crit} values, estimates derived from linear regressions with a negative slope, negative y-intercept, or $r^2 < 0.3$ were excluded. A 5-minute interval was chosen to optimize linear regression performance while maintaining temporal resolution based on our dataset's sampling frequency. Linear interpolation was used to impute missing P_{crit} values. TPP was then estimated as the difference between average MAP and the estimated P_{crit} value for each interval.

Definition of AKI

AKI was defined according to the creatinine-based criteria used in the Kidney Disease: Improving Global Outcomes (KDIGO) definition.¹⁸ Baseline creatinine was defined as the preoperative creatinine level measured closest to the time of surgery. Creatinine values obtained within 72 hours postoperatively were compared with the baseline values to identify AKI cases. The urine output criteria of the KDIGO definition were not used because they have been shown to be overly sensitive and nonspecific after cardiac surgery and may be confounded by early diuretic administration.^{19,20}

Unsupervised Machine Learning for Hemodynamic Phenotype Identification

To further characterize the relationship between TPP and AKI, postoperative hemodynamic phenotypes were identified by unsupervised machine learning with *k*-means clustering. Machine learning models were trained with a comprehensive feature set combining static and time-series hemodynamic and pharmacologic data. Static features included STS registry elements encompassing patient and operative variables. Hemodynamic and pharmacologic values were derived from the resampled time-series hemodynamic and VIS data.

All continuous variables were standardized as *z* scores. Variables with $>20\%$ missingness were excluded; otherwise, missing values were imputed as the median for continuous variables and the mode for categorical ones. Dimensionality was reduced by principal component analysis, and clustering was performed with *k*-means ($k = 2$); the optimal number of clusters was determined from the within-cluster sum of squares. Hemodynamic and clinical variables, including TPP, VIS, and AKI incidence, were then compared between clusters.

Statistical Analysis

Categorical variables are presented as number (percentage) and continuous variables as mean $\pm$ standard deviation or median (quartile 1-quartile 3), as

TABLE 1. Baseline characteristics of patients who did and did not have acute kidney injury (AKI)

Characteristic	No AKI (n = 1009)	AKI (n = 215)	P value*
Age (y)	66.0 (58.0-73.0)	68.0 (60.0-74.0)	.109
Male	713 (70.7)	145 (67.4)	.393
Left ventricular ejection fraction (%)	58.0 (51.2-61.0)	58.0 (47.0-61.0)	.311
Previous cardiac surgery	151 (15.0)	52 (24.2)	<.001
Medical history			
Hypertension	883 (87.5)	202 (94.0)	.010
Diabetes mellitus	361 (35.8)	99 (46.0)	.009
Heart failure	380 (37.7)	113 (52.6)	<.001
Chronic lung disease	198 (19.6)	57 (26.5)	.030
Liver disease	29 (2.9)	15 (7.0)	.006
Arrhythmia	260 (25.8)	79 (36.7)	<.001
Cerebrovascular disease	209 (20.7)	54 (25.1)	.182
Stroke	117 (11.6)	31 (14.4)	.299
Peripheral vascular disease	152 (15.1)	46 (21.4)	.029
Operative characteristics			
Nonelective	473 (46.9)	135 (62.8)	<.001
Coronary artery bypass grafting	490 (48.6)	85 (39.5)	.002
Aortic valve replacement	86 (8.5)	21 (9.8)	.650
Mitral valve replacement	164 (16.3)	24 (11.2)	.076
Aortic surgery	200 (19.8)	69 (32.1)	<.001
Other operation†	69 (6.8)	16 (7.4)	.864
Cardiopulmonary bypass time (min)	121.0 (94.0-155.0)	139.0 (97.0-192.8)	<.001
Crossclamp time (min)	80.0 (57.0-108.0)	86.0 (60.2-132.5)	.005
Lowest intraoperative temperature (°C)	31.5 (29.6-33.5)	30.1 (27.2-33.4)	<.001
Lowest intraoperative hemoglobin level (g/dL)	8.6 (7.4-10.1)	7.6 (6.8-8.6)	<.001
Highest intraoperative glucose level (mg/dL)	218.0 (182.8-257.0)	235.0 (192.5-268.0)	.001
Intraoperative blood products	437 (43.3)	154 (71.6)	<.001

Values are presented as median (quartile 1-quartile 3) or n (%). *Wilcoxon rank-sum test, χ^2 test, or Fisher exact test, as appropriate. †Other operation includes combined procedures.

appropriate. Categorical comparisons were performed by using the χ^2 test. For continuous variables, the Wilcoxon rank-sum test was used to assess between-groups differences in univariate analysis.

The relationship was assessed between AKI and postoperative outcomes, including mortality and major morbidity (a composite outcome comprising postoperative renal failure, stroke, prolonged ventilation, sternal wound infection, and reoperation). Mean perfusion parameters were compared between patients with and without AKI, using averages over the first 24 hours postoperatively. Logistic regression was used to identify optimal TPP thresholds for stratifying AKI risk. To adjust for operative and patient-level confounders, a propensity score analysis was performed using overlap weighting based on 19 covariates: a priori identified factors (surgical operation and preoperative creatinine) and 14 additional variables that differed significantly between the AKI and non-AKI cohorts.²¹ These weights were incorporated into logistic regression models to compare the association between TPP and AKI. Subgroup analyses assessed TPP and MAP stratified by VIS. All analyses were conducted with Python version 3.11.8.

RESULTS

Patient Characteristics

Among the 1224 patients (median age, 66 years [58-73 years]; 70% men; median baseline creatinine level, 0.97 mg/dL [0.82-1.21 mg/dL]) included in the study, 215 (17.6%) developed AKI within 72 hours after surgery. The distribution of AKI by stage was 14.3% (n = 166), 2.8% (n = 32), and 1.5%

(n = 17) for stages 1, 2, and 3, respectively. Compared with patients without AKI, those who developed AKI had higher baseline creatinine values (1.1 mg/dL [0.9-1.4 mg/dL] vs 0.8 mg/dL [0.8-1.2 mg/dL]; $P < .001$) and a greater burden of comorbidities (Table 1). Patients with postoperative AKI less often had coronary artery bypass grafting (39.5% vs 48.6%; $P = .02$) and more often had aortic surgery (32.1% vs 19.8%; $P < .001$). Additionally, patients with AKI had more complicated operative courses, with longer cardiopulmonary bypass times and transfusion requirements (Table 1).

TABLE 2. Postoperative adverse events in patients who did and did not have acute kidney injury (AKI)

Outcome*	No AKI	AKI	P value†
In-hospital mortality	20 (2.0)	34 (15.8)	<.001
Stroke	18 (1.8)	20 (9.3)	<.001
Sternal wound infection	15 (1.6)	5 (2.3)	.396
Sepsis	16 (1.6)	15 (7.0)	<.001
Reoperation	34 (3.4)	21 (9.8)	<.001
Prolonged ventilation	110 (10.9)	88 (40.9)	<.001

Values are presented as n (%). *Defined according to Society of Thoracic Surgeons criteria for postoperative complications. † χ^2 or Fisher exact test, as appropriate.

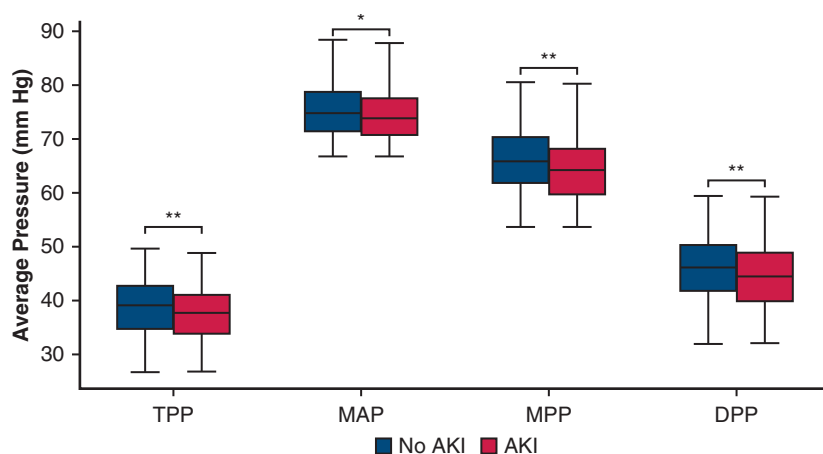


FIGURE 1. Comparison of perfusion measures by acute kidney injury (AKI) status. Horizontal bar represents the median; the upper and lower borders of the box represent the 75th and 25th quartiles, respectively. Whiskers represent the minimum and maximum values. TPP, Tissue perfusion pressure; MAP, mean arterial pressure; MPP, mean perfusion pressure; DPP, diastolic perfusion pressure. * $P < .05$. ** $P < .005$.

Patients who developed AKI had significantly worse outcomes than those who did not. These differences included higher rates of in-hospital mortality (15.8% vs 2.0%; $P < .001$) and major morbidity (45.1% vs 14.2%; $P < .001$), including nearly all STS-defined major morbidities (Table 2). Postoperative dialysis was required in 24.2% ($n = 52$) of the AKI cohort, and patients with severe AKI (stage 2 or 3) had higher in-hospital mortality than patients with stage 1 AKI (28.6% vs 12.0%; $P = .010$).

TPP

The overall median TPP for the cohort was 38.5 mm Hg (33.5–42.9 mm Hg); the median MAP was 74.9 mm Hg (70.9–79.9 mm Hg). The median MPP and DPP were 65.7 mm Hg (60.3–71.5 mm Hg) and 45.8 mm Hg (40.3–50.9 mm Hg), respectively. Patients who developed AKI had significantly lower perfusion values than patients without AKI across all 4 perfusion measures: TPP (37.6 mm Hg [33.7–41.0 mm Hg] vs 39.0 mm Hg [34.6–42.6 mm Hg]; $P < .001$), MAP (73.8 mm Hg [70.6–77.6 mm Hg] vs 74.8 mm Hg [71.4–78.7 mm Hg]; $P = .017$), MPP (64.2 mm Hg [59.7–68.2 mm Hg] vs 65.8 mm Hg [61.8–70.3 mm Hg]; $P < .001$), and DPP 44.4 mm Hg [39.8–48.9 mm Hg] versus 46.0 mm Hg [41.8–50.2 mm Hg]; $P < .001$) (Figure 1). Subset analysis revealed similar trends when patients were stratified by operation type (Figure E2) and AKI severity (Figure E3).

Logistic regression analysis identified 38 mm Hg as the optimal TPP threshold for differentiating AKI risk (odds ratio, 1.75; 95% CI, 1.30–2.35). Patients with average TPP values below this threshold had significantly higher rates of AKI than those with values above it (20.1% vs 13.3%; $P = .001$) (Figure 2). Average TPP <38 mm Hg remained an independent predictor of AKI even after adjustment for confounding operative and patient-level risk factors (odds ratio, 1.81; 95% CI, 1.33–2.48).

TPP AND VIS

Among the entire cohort, 812 patients (66.3%) required vasoactive medications for more than 1 hour, and 102 (8.3%) required high doses (maximum VIS >15). Patients requiring vasoactive medications had lower average TPP (37.4 vs 39.7 mm Hg; $P < .001$) and MAP (75.3 vs 78.3 mm Hg; $P < .001$). A stepwise increase in the incidence of AKI and a decrease in average TPP was associated with higher VIS scores (Figure 3). Subset analysis of perfusion pressures by VIS score revealed a significant decline in average TPP but no change in MAP at higher VIS scores. For patients requiring vasoactive medications, average TPP <38 mm Hg was associated with a greater risk of AKI

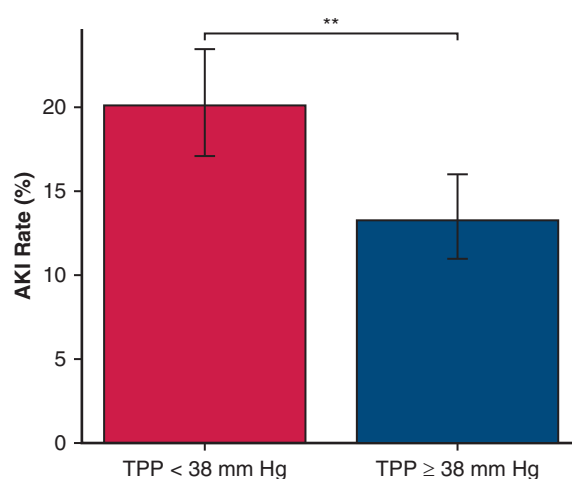


FIGURE 2. Comparison of acute kidney injury rates for patients below and above the tissue perfusion pressure (TPP) threshold of 38 mm Hg. Horizontal bar represents the median; the upper and lower borders of the box represent the 75th and 25th quartiles, respectively. Whiskers represent the minimum and maximum values. ** $P < .005$.

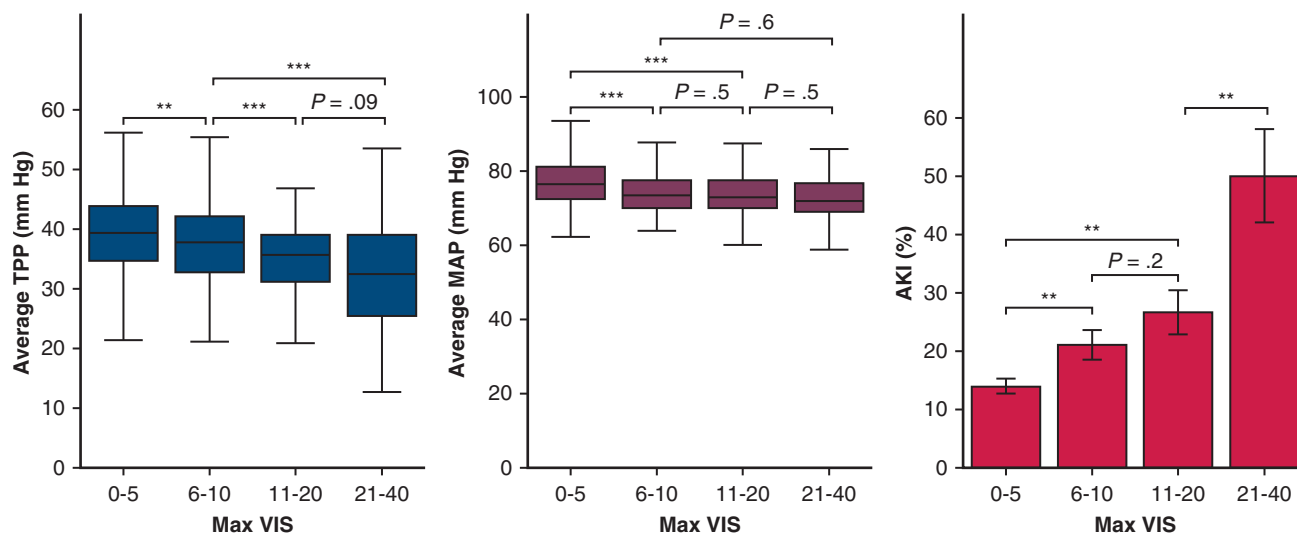


FIGURE 3. Comparison of (A) average tissue perfusion pressure (TPP), (B) average mean arterial pressure (MAP), and (C) incidence of acute kidney injury (AKI) across different maximum vasoactive inotrope scores (VISs). Horizontal bar represents the median; the upper and lower borders of the box represent the 75th and 25th quartiles, respectively. Whiskers represent the minimum and maximum values.

independent of average MAP (adjusted odds ratio, 1.69; 95% CI, 1.17-2.45).

Hemodynamic Phenotypes

The *k*-means clustering algorithm identified 2 distinct clusters of patients with different hemodynamic phenotypes after cardiac surgery. Cluster 1 appeared to have greater hemodynamic instability (Figure 4), including lower TPP (35.6 mm Hg [30.9-39.8 mm Hg] vs 40.4 mm Hg [35.9-44.5 mm Hg]; $P < .001$), than Cluster 2. Moreover, Cluster 1 patients had higher average VIS (2.8 [0.3-6.8] vs 0.5 [0.0-2.5]; $P < .001$) and CVP (10.9 mm Hg [8.4-13.1 mm Hg] vs 8.1 mm Hg [6.0-10.2 mm Hg]; $P < .001$). Patients in Cluster 1 were higher-risk operative candidates, with a greater burden of comorbidities (Table 3) and worse preoperative cardiac function, having higher rates of heart failure (57.9% vs 29.6%; $P < .001$), lower ejection fractions (55.8% [41.0%-60.0%] vs 58.4% [53.0%-61.8%]; $P < .001$), and a higher incidence of preoperative cardiogenic shock (8.4% vs 0.7%, $P < .001$). Thus, these patients had worse outcomes, including higher incidences of AKI (29.8% vs 10.1%; $P < .001$), major morbidity (33.0% vs 10.7%; $P < .001$), and in-hospital mortality (10.0% vs 1.1%; $P < .001$).

DISCUSSION

In this study, we used routinely collected EMR data to evaluate associations among several hemodynamic indices, including the novel parameter TPP, and postoperative AKI in cardiac surgery patients. We found that patients who developed AKI had significantly lower renal perfusion according to all four measures analyzed. Our key findings are that lower average TPP in the first 24 postoperative hours was associated with higher AKI incidence, a TPP

threshold of 38 mm Hg most effectively stratified AKI risk, and TPP was predictive of AKI independent of MAP in patients requiring vasoactive medications.

These findings contribute to the broader effort to reduce cardiac surgery–associated AKI, a complication we found to be associated with worse postoperative outcomes. Although KDIGO guidelines recommend hemodynamic optimization as part of AKI prevention, they do not specify ideal perfusion targets, largely because strong data on optimal perfusion thresholds are lacking.²² Most existing studies have derived blood pressure targets from septic shock populations, with mixed results regarding optimal MAP thresholds for AKI prevention.^{8,23-25} MAP alone may not adequately reflect renal perfusion, especially in patients with elevated CVP or altered vascular tone.

Our results align with those of prior studies suggesting that perfusion metrics incorporating CVP, such as MPP and DPP, provide more detail regarding renal perfusion than MAP alone.²⁶⁻²⁸ Furthermore, our analysis suggests that TPP, which accounts for the influence of vasomotor tone on tissue perfusion, offers additional insight, especially in shock states common in the postoperative period. This builds on prior work by Chandrasekhar and colleagues,¹⁷ who developed the method for estimating TPP from EMR data and linked this measure to global outcomes like mortality and length of stay. Our study extends these findings to renal outcomes, identifying a potential TPP target threshold for mitigating AKI.

Of particular relevance is our observation that TPP was predictive of AKI in patients requiring vasoactive medications, even after adjustment for MAP. It is possible that in these patients, standard metrics such as MAP obscure underlying perfusion deficits, whereas TPP may better reflect

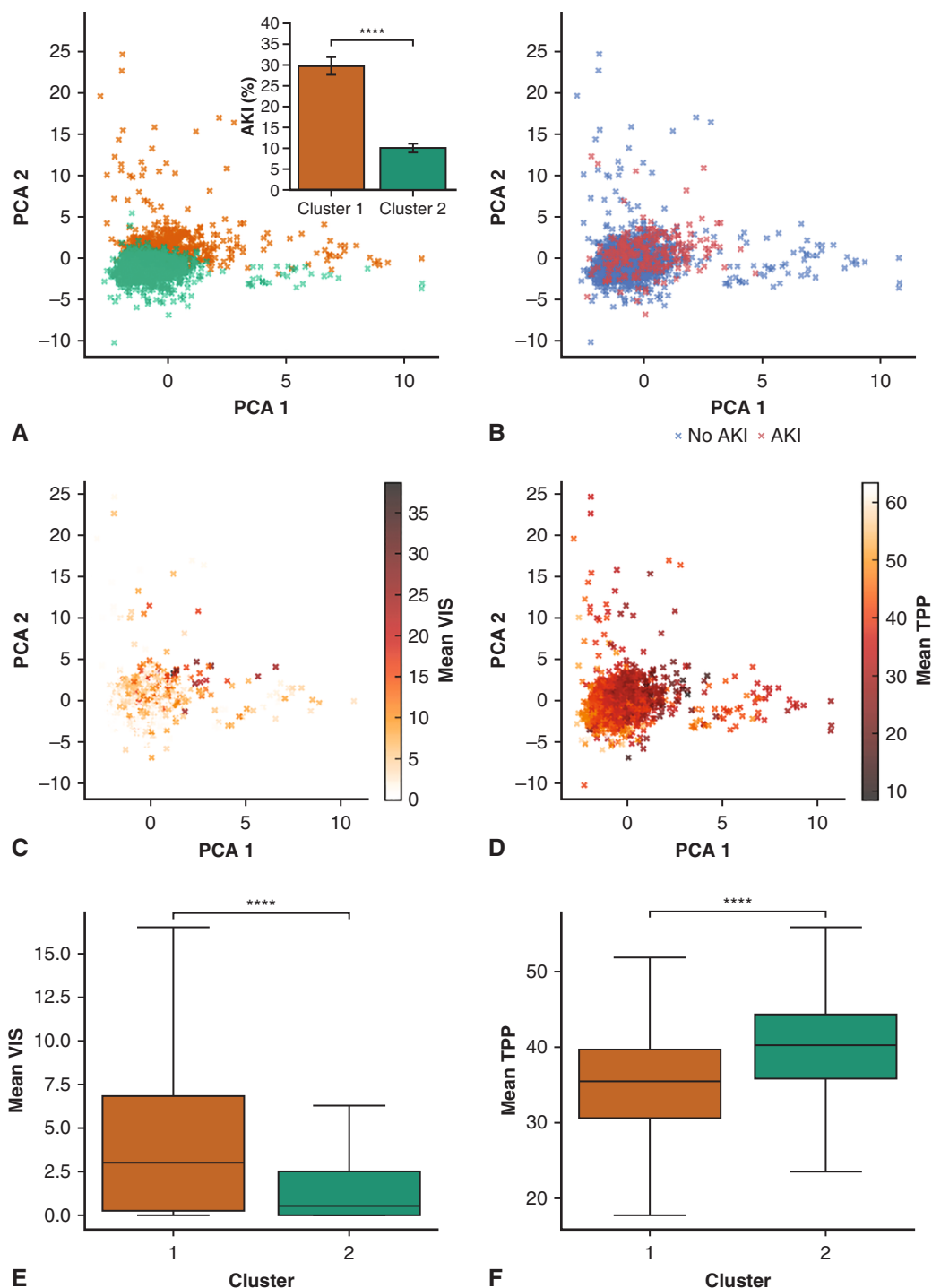


FIGURE 4. Hemodynamic phenotypes derived from unsupervised machine learning analysis. Principal component analysis (PCA) visualizations annotated by cluster (A), acute kidney injury (AKI) status (B), mean vasoactive inotrope score (VIS) (C), and average tissue perfusion pressure (TPP) (D). Box-plots show differences in average VIS (E) and TPP (F) by cluster.

true renal perfusion—specifically, glomerular perfusion pressure, particularly in patients with hemodynamic instability. Because TPP integrates both macrocirculatory (ie, MAP) and microcirculatory measurements (ie, P_{crit}), it

may characterize renal perfusion more comprehensively. Although hypothesis-generating, this analysis suggests that TPP could serve as an additional guide for postoperative hemodynamic management.

TABLE 3. Baseline characteristics associated with 2 distinct hemodynamic phenotypes identified with *k*-means clustering analysis

Characteristic	Cluster 1 (n = 451)	Cluster 2 (n = 741)	P value
Age (y)	68.0 (60.0-74.0)	65.0 (57.0-72.0)	<.001
Male	296 (65.6)	538 (72.6)	.013
Left ventricular ejection fraction	55.8 (41.0-60.0)	58.4 (53.0-61.8)	<.001
Previous cardiac surgery	85 (18.8)	90 (12.1)	.002
Medical history			
Hypertension	412 (91.4)	645 (87.0)	.029
Diabetes mellitus	229 (50.8)	225 (30.4)	<.001
Heart failure	261 (57.9)	219 (29.6)	<.001
Chronic lung disease	115 (25.5)	128 (17.3)	.001
Liver disease	18 (4.0)	22 (3.0)	.433
Arrhythmia	170 (37.7)	151 (20.4)	<.001
Cerebrovascular disease	114 (25.3)	141 (19.0)	.013
Stroke	63 (14.0)	80 (10.8)	.123
Peripheral vascular disease	77 (17.1)	108 (14.6)	.283
Operative characteristics			
Nonelective	351 (77.8)	242 (32.7)	<.001
Coronary artery bypass grafting	37 (8.2)	70 (9.4)	.999
Aortic valve replacement	86 (8.5)	21 (9.8)	.533
Mitral valve replacement	63 (14.0)	120 (16.2)	.342
Aortic surgery	89 (19.7)	156 (21.1)	.637
Cardiopulmonary bypass time (min)	128.0 (101.0-172.0)	119.0 (90.2-150.0)	<.001
Crossclamp time (min)	84.5 (62.0-120.0)	79.0 (56.0-106.2)	.001
Lowest intraoperative temperature (°C)	31.2 (28.2-33.3)	31.4 (29.6-33.6)	.145
Lowest intraoperative hemoglobin level (g/dL)	7.6 (6.8-8.7)	9.0 (7.8-10.4)	<.001
Highest intraoperative glucose level (mg/dL)	237.0 (203.2-273.0)	211.0 (175.0-248.5)	<.001
Intraoperative blood products	299 (66.3)	272 (36.7)	<.001

These findings have practical implications for postoperative care after cardiac surgery. First, MAP alone is probably insufficient for assessing renal perfusion, particularly in patients with abnormal CVP or high vasopressor requirements. Second, incorporating TPP into routine hemodynamic assessment could enhance AKI risk stratification, with a threshold of 38 mm Hg representing a potential target for goal-directed therapy. Third, because TPP can be derived from standard EMR data, its future integration into clinical decision-support systems could provide real-time perfusion assessments to assist hemodynamic management.

This study has several limitations. First, it relies on retrospective data from a single center, which may limit generalizability. Residual confounders may remain despite adjustment for operation and patient-level risk factors using propensity scores. Prospective studies are needed to determine whether TPP-guided management improves renal outcomes after cardiac surgery. Second, while TPP and P_{crit} were estimated by a validated modeling approach, this methodology depends on specific physiologic assumptions and EMR data quality. Using rigorous inclusion criteria to ensure physiologically plausible P_{crit} estimates resulted in 43.1% of estimates being excluded, probably due to limitations in sampling frequency

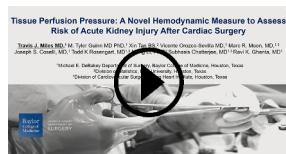
and measurement artifact. Nonetheless, we observed a consistent association between lower TPP and AKI. Finally, our study focuses primarily on the influence of postoperative renal perfusion on AKI. Although we attempted to adjust for preoperative and intraoperative risk factors, intraoperative renal insults contributing to AKI were not completely accounted for. Nonetheless, the early postoperative period represents a critical opportunity to optimize renal outcomes through hemodynamic management. In this context, we believe TPP remains a clinically meaningful target for mitigating AKI.

CONCLUSIONS

Lower TPP in the early postoperative period may be associated with greater risk of postoperative AKI after cardiac surgery. We identified a potential threshold value for TPP that could aid in risk stratification and found that TPP may be particularly relevant in patients receiving higher doses of vasoactive medications. These findings suggest that TPP could be a useful adjunct to traditional hemodynamic parameters in assessing renal perfusion, and they support further investigation into TPP's potential value in guiding postoperative management.

Webcast

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Audio

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Conflict of Interest Statement

Dr Coselli participates in clinical studies with and/or consults for Terumo Aortic, Medtronic, W.L. Gore & Associates, CytoSorbents, Edwards Lifesciences, and Abbott Laboratories and receives royalties and grant support from TerumoAortic. Dr Moon is a consultant/advisory board member for Medtronic and Edwards Lifesciences. Dr Chatterjee has served on advisory boards for Edwards Lifesciences, La Jolla Pharmaceutical Company, Eagle Pharmaceuticals, and Baxter Pharmaceuticals. All other authors reported no conflicts of interest.

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Stephen N. Palmer, PhD, ELS, of the Department of Scientific Publications at The Texas Heart Institute contributed to the editing of the manuscript. Decisio Health (Houston, Tex) assisted with data acquisition.

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Key Words: acute kidney injury, tissue perfusion pressure, cardiac surgery

APPENDIX E1. DATA EXTRACTION AND PROCESSING

Postoperative critical care data were extracted from the Baylor College of Medicine instance of the Epic EHR (Epic Systems Corporation) by an in-house team of surgeons and biomedical informaticists as previously described.^{E1} Briefly, record-level data extracts of fields containing hemodynamic and laboratory data were queried from the Clarity database by using Structured Query Language. This dataset was then supplemented with high-frequency sampled blood pressure and heart rate data collected from bedside telemetry monitors and available through a third-party digital health company (Decisio Health).

Extracted electronic health record data were uploaded into the open-source Python software version 3.11.8 (Python Software Foundation), in which hemodynamic and pharmacologic features of interest were systematically cleaned and transformed into time series format, with time 0 defined as time of intensive care unit admission. Data validity was ensured by removing extreme outliers (>99th percentile and <first percentile) and physiologically infeasible measurements (eg, systemic blood pressure >300 mm Hg), as is common in analyses involving electronic health record data.^{E1,E2}

Vasoactive Inotrope Score

The vasoactive inotrope score (VIS) was used to quantify total dose of vasopressors and inotropes in the 24 hours after cardiac surgery. The number of vasoactive medications and doses of individual vasoactive medications are poor surrogates for shock severity, so the concept of VIS was developed to standardize nomenclature and quantify the overall requirement for vasoactive medications. The total VIS is calculated by converting vasopressor and inotrope doses into equivalency doses and summing them, thus providing a standardized metric to quantify vasopressor and inotrope requirements. Formula for calculating VIS.^{E3-E6}

$$\text{VIS} = \text{dopamine} + \text{dobutamine} + 100 \times (\text{epinephrine} + \text{norepinephrine} + \text{phenylephrine}) + 10 \times \text{milrinone} + 10 \times \text{vasopressin (U/min)} + 10 \times \text{angiotensin-II } (\mu\text{g/kg/min}).$$

This aggregate score reflects the intensity of pharmacologic support but does not differentiate between vasodilatory and vasoconstrictive components. As such, whereas VIS provides a practical summary of hemodynamic support burden, it does not capture the precise physiologic effects of individual agents on microvascular perfusion, afterload, or systemic vascular resistance. Nonetheless, it remains among the more commonly reported measures of hemodynamic support in the cardiac surgery literature.

The influences of vasoactive therapy on tissue perfusion pressure (TPP) are complex and incompletely understood. For example, agents like norepinephrine can increase mean arterial pressure while also increasing critical closing pressure via microvascular vasoconstriction, potentially resulting in an unchanged or even reduced TPP. Conversely, inodilators can improve cardiac output and reduce critical closing pressure without significantly increasing mean arterial pressure. Thus, the interplay between vasoactive medications and TPP is multifaceted and may reflect underlying vasoplegia, low cardiac output states, or both. The characterization of the effect of various pharmacologic or clinical interventions on TPP represents a promising area of future research but likely requires study in a prospective fashion.

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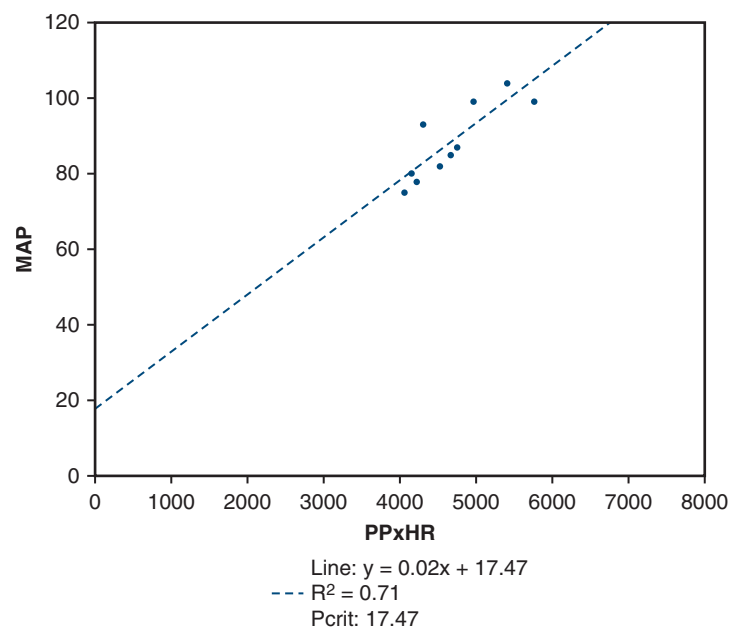


FIGURE E1. Method for estimating the critical closing pressure (Pcrit) from routinely collected vital sign data. MAP, Mean arterial pressure; PP×HR, product of pulse pressure and heart rate.

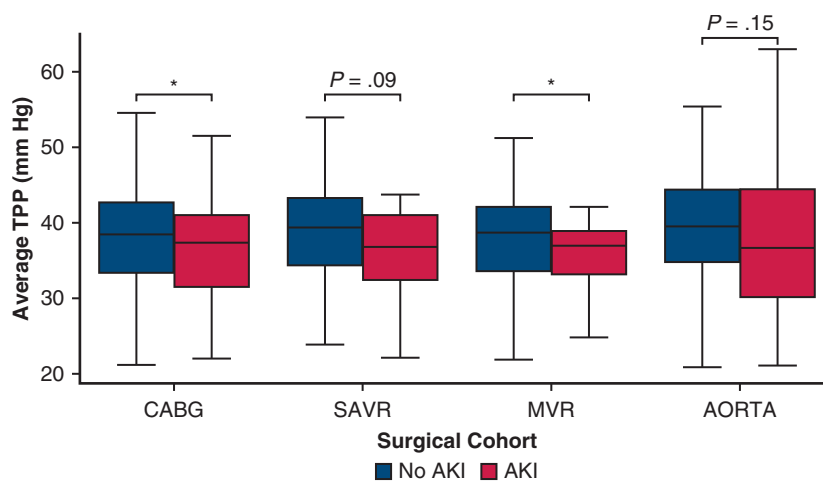


FIGURE E2. Comparison of average tissue perfusion pressure (TPP) by acute kidney injury (AKI) status for the 4 principal surgical procedures. Horizontal bar represents the median; the upper and lower borders of the box represent the 75th and 25th quartiles, respectively. Whiskers represent the minimum and maximum values. CABG, Coronary artery bypass grafting; SAVR, surgical aortic valve replacement; MVR, mitral valve replacement. $*P < .05$.

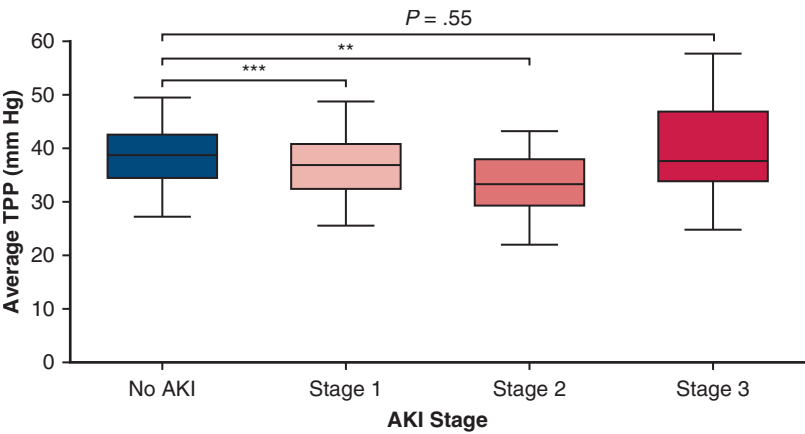


FIGURE E3. Comparison of average tissue perfusion pressure (TPP) by acute kidney injury (AKI) severity according to Kidney Disease Improve Global Outcomes stage. ***P* < .005. ****P* < .0005.

TABLE E1. Converting vasoactive medications to vasoactive inotrope score (VIS)

Medication	Dose	VIS contribution
Dopamine	1 μg/kg/min	+1 × dose
Dobutamine	1 μg/kg/min	+1 × dose
Epinephrine	0.01 μg/kg/min	+100 × dose
Norepinephrine	0.01 μg/kg/min	+100 × dose
Phenylephrine	0.01 μg/kg/min	+100 × dose
Milrinone	0.1 μg/kg/min	+10 × dose
Vasopressin	0.01 U/min	+10 × dose (in U/min)
Angiotensin II	10 ng/kg/min (0.01 μg/kg/min)	+10 × dose (in μg/kg/min)